MDPI

Article

From Cain to Matignon: Exact Positive-Diagonal Stability in Dimension Three and a Nonlinear Caputo Ecological Realization

Ibrahim Alraddadi [1,*]

[1] Department of Mathematics, Faculty of Science, Islamic University of Madinah, Madinah 42351, Saudi Arabia
* Correspondence: ialraddadi@iu.edu.sa

Abstract

Positive-diagonal D-stability asks whether stability persists under every positive diagonal left scaling. We replace the classical Hurwitz half-plane by the Matignon sector for commensurate Caputo systems and solve the resulting real three-dimensional problem on the full-dimensional strict-$P(-A)$ stratum. Four scale-invariant quantities $(\beta_{12}, \beta_{13}, \beta_{23}, \kappa)$ reduce the complete positive-diagonal orbit to an open simplex, and for $2/3 < \alpha < 1$ we prove the exact criterion $\kappa < T_\alpha(\beta)$. At $\alpha = 1$, T_α collapses to Cain's classical threshold, while $T_\alpha > T_1$ for every $\alpha < 1$, producing an explicit genuinely fractional band. The threshold objective is strictly convex in logit coordinates, has a unique nondegenerate optimizer, and yields the minimum-dimension result: three is the minimum dimension in which the genuinely fractional multiplicative-D-stable class has nonempty full-dimensional interior. We then construct a nonlinear three-species Caputo Kolmogorov realization with double-Allee basal growth and intraguild predation. A natural competing-consumer architecture is proved unable to enter the fractional-only band, whereas the intraguild-predation architecture realizes an open biological subset of it. Varying only the Allee threshold moves the fixed biological system transversally across Cain's classical D-stability boundary into the fractional-only Matignon region. A reproducible integration of the full nonlinear model at a certified orbit point displays decaying Caputo transients at $\alpha = 0.9$, while the same parameters, rate scaling and initial perturbation leave a fixed local neighborhood at $\alpha = 1$.

Keywords: fractional-order systems; Caputo derivative; D-stability; Matignon stability; P-matrix; Cain criterion; intraguild predation; double Allee effect; ecological stability

MSC: 15A18; 34A08; 34D20; 92D25

1. Introduction

Stability under a single Jacobian is often too weak a robustness requirement. In matrix D-stability one asks instead whether the spectrum remains in a prescribed stability region after *every* positive diagonal left scaling. For the classical Hurwitz half-plane this problem has a long history, with exact low-dimensional results, interior and strong-D-stability theory, and structural criteria based on principal minors and feedback cycles [1–3]. The question addressed here is what becomes of this all-diagonal robustness requirement when the underlying dynamics are commensurate fractional-order systems.

For a Caputo system of order $0 < \alpha < 1$, asymptotic stability of the linearization is governed by the Matignon sector rather than the open left half-plane: every eigenvalue must satisfy

$$|\arg \lambda| > \frac{\alpha \pi}{2}.$$

Received:
Accepted:
Published:

Copyright: © 2026 by the authors. Submitted to *Mathematics* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license.

We use the original proceedings formulation together with a modern published treatment of Caputo stability [4,5]. For $\alpha < 1$, the Matignon region strictly contains the left half-plane, so an equilibrium may be fractionally stable even when a conjugate pair has positive real part. This familiar fixed-matrix phenomenon is not our novelty; it is the geometric reason a fractional analogue of multiplicative D-stability can differ from the classical one.

Generalized matrix stability with respect to spectral regions and multiplier classes is already established. In particular, Kushel's unifying framework includes multiplicative positive-diagonal actions, and Kushel and Pavani give forbidden-boundary criteria for generalized D-stability in unbounded LMI regions, including conic sectors and their complements [3,6]. Thus the problem considered here is *not* the first necessary-and-sufficient formulation of generalized D-stability. Our contribution is lower-dimensional and explicit: for real 3×3 matrices on the full-dimensional strict-$P(-A)$ stratum, we eliminate the entire positive-diagonal multiplier orbit and reduce the Matignon condition to a scalar threshold in four invariant coordinates.

The classical endpoint of this construction is equally important. Cain's exact criterion for real 3×3 D-stable matrices already shows that principal minors, homogeneity in the diagonal multiplier, and an exact minimization over the multiplier orbit are classical ingredients [1]. The present threshold should therefore be read as a fractional deformation of Cain's boundary, not as an unrelated sufficient test. Other adjacent theories are also occupied: P-matrix eigenvalue wedges [7], fixed-polynomial fractional Routh–Hurwitz criteria [8,9], and structured cyclic fractional-network conditions [10]. The exact arbitrary-β all-diagonal threshold derived below is not implied by any of these results.

The first structural result is dimensional. In dimension two we obtain the exact classification

$$A \in \mathcal{F}_\alpha^{(2)} \iff \det A > 0, \qquad a_{11} \le 0, \qquad a_{22} \le 0,$$

and the genuinely fractional difference from classical D-stability lies only on a codimension-two diagonal-equality set. Hence it has empty full-dimensional interior. In dimension three the situation changes qualitatively. On the strict-$P(-A)$ stratum, four normalized principal-minor invariants

$$(\beta_{12}, \beta_{13}, \beta_{23}, \kappa)$$

encode the positive-diagonal orbit. For $2/3 < \alpha < 1$ we prove

$$A \in \mathcal{F}_\alpha^{(3)} \iff \kappa < T_\alpha(\beta),$$

where T_α is an exact two-dimensional simplex minimum. At $\alpha = 1$,

$$T_1(\beta) = \left(\sqrt{\beta_{12}} + \sqrt{\beta_{13}} + \sqrt{\beta_{23}}\right)^2,$$

which is Cain's classical boundary. For every $2/3 < \alpha < 1$,

$$T_\alpha(\beta) > T_1(\beta),$$

so

$$T_1(\beta) < \kappa < T_\alpha(\beta)$$

is an open, genuinely fractional, all-positive-diagonal stability band. For $0 < \alpha \le 2/3$, Kellogg's P-matrix wedge gives the complementary low-order characterization. Together with the exact two-dimensional obstruction, this yields

$$\min\left\{n : \operatorname{int}\left(\mathcal{F}_\alpha^{(n)} \setminus \mathcal{D}_H^{(n)}\right) \neq \varnothing\right\} = 3 \qquad \forall\, 0 < \alpha < 1.$$

The threshold is not merely an implicit minimum. Its logarithm is globally strictly convex in logit coordinates, giving a unique nondegenerate optimizer and smooth parameter dependence. It decreases strictly with α and approaches Cain's surface linearly as $\alpha \to 1^-$; Siami's cyclic condition is recovered as the fully symmetric slice $\beta_{12} = \beta_{13} = \beta_{23} = 1$ [10].

We next ask whether the exact difference band is only an abstract matrix phenomenon. Recent published analyses examine stability and nonlinear dynamics in commensurate and incommensurate fractional ecological systems, including memory-sensitive predator–prey stability, positivity, and bifurcation [11–14]. These studies concern specific models or prescribed parameters and do not yield an exact *all-positive-diagonal* spectral classification for arbitrary strict-P real three-by-three matrices. The novelty here is the exact elimination of the diagonal multiplier into a four-invariant necessary-and-sufficient threshold, plus its unique-optimizer geometry and constructive nonlinear realization.

Classical ecological D-stability is established [15,16]; IGP and IGP with Allee effects are standard motifs [17,18], fractional IGP already exists [19], and fractional double-Allee predator–prey models are known [20,21]. Accordingly, novelty is claimed only for the exact all-diagonal realization results below, not for the ecological ingredients themselves.

The ecological contribution is instead an exact realization theorem. We first prove that the most natural double-Allee prey with two competing consumers is structurally incapable of entering the open fractional-only band: its directed three-cycle contribution has the wrong sign and forces $\kappa < T_1(\beta)$ throughout the strict-P region. Replacing inter-consumer competition by an IGP/omnivory link reverses one directed three-cycle and creates a four-dimensional open invariant family. We then embed any required positive prey self-slope into a double-Allee growth law at a positive coexistence equilibrium, proving that for every fixed $0 < \alpha < 1$ there is a nonempty open set in the full biological parameter space for which

$$J \in \mathcal{F}_\alpha^{(3)} \setminus \mathcal{D}_H^{(3)}.$$

Finally, the Allee threshold itself becomes a control coordinate: along the positive strict-P coexistence branch, increasing the threshold decreases both the prey equilibrium X and its effective restoring slope $s = -g'_{\mathrm{DA}}(X)$, so $t = 1/s$ increases monotonically. The resulting invariant path crosses Cain's boundary at most once and, when tuned to cross, does so transversally. Thus varying only the Allee threshold can carry one fixed biological system from classical D-stability into the fractional-only all-diagonal Matignon region.

The contributions can be summarized as follows:

1. an exact two-dimensional obstruction and the minimum-dimension theorem for nonempty full-dimensional genuinely fractional multiplicative-D-stability;
2. an exact real 3×3 four-invariant characterization of positive-diagonal Matignon stability, recovering Cain at $\alpha = 1$;
3. strict convexity, uniqueness and asymptotic geometry of the threshold T_α;
4. a structural no-go theorem for a natural competing-consumer double-Allee architecture;
5. a constructive IGP realization of an open four-invariant family and an open biological fractional-only region for every $0 < \alpha < 1$;
6. a rigorous Allee-threshold mechanism that crosses the exact classical Cain boundary while all other biological parameters remain fixed;
7. a reproducible full nonlinear time-domain comparison at a certified fractional-only orbit point.

The paper is organized accordingly. Section 2 introduces the positive-diagonal Matignon framework and the simplex normalization. Section 3 treats dimension two. Sections 4 and 5 give the exact three-dimensional characterization and threshold geometry.

Section 6 develops the ecological no-go, realization and Allee crossing. Section 7 records a certified interior realization and its nonlinear time-domain case study, and Section 8 discusses interpretation, limitations and higher-dimensional directions.

2. Positive-Diagonal Matignon Framework

Fix $0 < \alpha < 1$ and define

$$\Sigma_\alpha := \Big\{ z \in \mathbb{C} \setminus \{0\} : |\arg z| > \frac{\alpha\pi}{2} \Big\}. \tag{1}$$

For real $n \times n$ matrices,

$$\mathcal{F}_\alpha^{(n)} := \big\{ A \in \mathbb{R}^{n\times n} : \operatorname{spec}(DA) \subset \Sigma_\alpha \text{ for every positive diagonal } D \big\}, \tag{2}$$

and let $\mathcal{D}_H^{(n)}$ denote classical Hurwitz D-stability. The genuinely fractional difference class is

$$\mathcal{P}_\alpha^{(n)} := \mathcal{F}_\alpha^{(n)} \setminus \mathcal{D}_H^{(n)}. \tag{3}$$

Here D-stability is used only in the multiplicative matrix sense, not in the distinct control-theoretic meaning of root placement in a prescribed D-region. Matignon's criterion identifies Σ_α as the asymptotic-stability region of a commensurate Caputo linear system [4,5].

2.1. Projective Coordinates of the Positive-Diagonal Orbit

Suppose $-A$ is a strict P-matrix. Write $p_i = -a_{ii} > 0$ and $m_I = (-1)^{|I|} \det A[I] > 0$. For $D = \operatorname{diag}(d_1, \dots, d_n) \succ 0$, the characteristic coefficients, projective multiplier coordinates, and normalized principal-minor invariants are

$$c_k(D) = \sum_{|I|=k} m_I \prod_{i\in I} d_i, \qquad x_i = \frac{p_i d_i}{\sum_j p_j d_j}, \qquad \sum_i x_i = 1, \tag{4}$$

$$\beta_I = \frac{m_I}{\prod_{i\in I} p_i} \qquad (|I| \geq 2). \tag{5}$$

Thus the positive-diagonal cone modulo common scale is exactly the open simplex. If $a_D = \sum_i p_i d_i$ and the spectral variable is scaled by a_D, the orbit becomes

$$z^n + z^{n-1} + B_2(x) z^{n-2} + \cdots + B_n(x), \qquad B_k(x) = \sum_{|I|=k} \beta_I \prod_{i\in I} x_i. \tag{6}$$

This projective reduction is structural rather than a novelty claim; the principal-minor representation is classical and underlies feedback-cycle descriptions [22].

For $n = 3$, only four nontrivial orbit invariants remain:

$$\boxed{\beta_{12} = \frac{m_{12}}{p_1 p_2}, \quad \beta_{13} = \frac{m_{13}}{p_1 p_3}, \quad \beta_{23} = \frac{m_{23}}{p_2 p_3}, \quad \kappa = \frac{-\det A}{p_1 p_2 p_3}.} \tag{7}$$

Section 4 shows that these four quantities are sufficient to eliminate the universal positive-diagonal multiplier from the real three-dimensional Matignon problem.

3. Dimension Two: Exact Obstruction

The two-dimensional problem is completely rigid. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad D = \operatorname{diag}(d_1, d_2) \succ 0.$$

Then

$$\operatorname{tr}(DA) = d_1 a + d_2 d, \qquad \det(DA) = d_1 d_2 \det A.$$

Theorem 1 (exact 2×2 classification). *For every real 2×2 matrix A and every $0 < \alpha < 1$,*

$$A \in \mathcal{F}_\alpha^{(2)} \iff \det A > 0, \qquad a \le 0, \qquad d \le 0. \tag{8}$$

Moreover,

$$\mathcal{P}_\alpha^{(2)} = \left\{ \begin{pmatrix} 0 & b \\ c & 0 \end{pmatrix} : bc < 0 \right\}, \tag{9}$$

and hence

$$\operatorname{int} \mathcal{P}_\alpha^{(2)} = \varnothing.$$

Proof. Assume first that $\det A > 0$ and $a, d \le 0$. For every $D \succ 0$,

$$\det(DA) > 0, \qquad \operatorname{tr}(DA) \le 0.$$

If the eigenvalues are real, both are negative unless the trace vanishes; if they are nonreal, they form a conjugate pair with nonpositive real part. Therefore every eigenvalue has argument magnitude at least $\pi/2$, and since $0 < \alpha < 1$,

$$\frac{\pi}{2} > \frac{\alpha \pi}{2}.$$

Thus $A \in \mathcal{F}_\alpha^{(2)}$.

Conversely, if $\det A \le 0$, some positive diagonal scaling has a nonnegative real eigenvalue, so Matignon stability fails. Suppose $\det A > 0$ but $a > 0$. Set $D_t = \operatorname{diag}(t, 1)$. Then

$$\operatorname{tr}(D_t A) = ta + d > 0$$

for all sufficiently large t, while

$$\operatorname{disc}(D_t A) = (ta + d)^2 - 4t \det A > 0$$

for all sufficiently large t. Hence $D_t A$ has two positive real eigenvalues for large t, contradicting membership in $\mathcal{F}_\alpha^{(2)}$. The case $d > 0$ is symmetric.

For classical Hurwitz D-stability, the same determinant condition is required together with

$$d_1 a + d_2 d < 0 \qquad \forall d_1, d_2 > 0,$$

which is equivalent to $a \le 0$, $d \le 0$, and not both zero. Therefore, the fractional-only difference occurs exactly when $a = d = 0$ and $\det A = -bc > 0$, giving Equation (9). This set has empty interior in $\mathbb{R}^{2\times 2}$. □

Figure 1 places this obstruction beside the three-dimensional geometry developed below. In two dimensions the fractional-only set is confined to an equality locus; in dimension three an open band separates the Matignon and Cain boundaries.

Theorem 1 is the lower-dimensional half of the minimum-dimension result. The remaining task is to show that dimension three contains full-dimensional open subsets of $\mathcal{P}_\alpha^{(3)}$ for every $0 < \alpha < 1$.

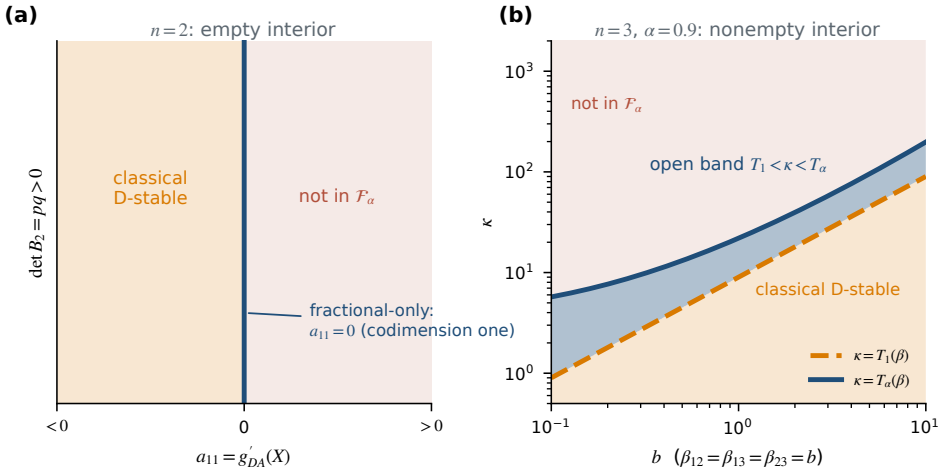


Figure 1. Dimension dependence of multiplicative positive-diagonal Matignon stability. In two dimensions, the difference from classical Hurwitz D-stability requires both diagonal matrix entries to vanish while the off-diagonal product is negative; this condition has empty full-dimensional interior. In dimension three, the classical Cain surface $\kappa = T_1(\beta)$ and fractional threshold $\kappa = T_\alpha(\beta)$ enclose an open fractional-only interval for the strict-$P(-A)$ class.

4. Exact Dimension-Three Characterization

We now specialize the simplex reduction to real 3×3 matrices. Assume first that $-A$ is a strict P-matrix and write

$$p_i = -a_{ii} > 0, \qquad m_{12}, m_{13}, m_{23} > 0, \qquad q = -\det A > 0,$$

with the normalized invariants in Equation (7). For

$$D = \operatorname{diag}(d_1, d_2, d_3) \succ 0$$

the characteristic polynomial is

$$\det(\lambda I - DA) = \lambda^3 + a_D \lambda^2 + b_D \lambda + c_D, \tag{10}$$

where

$$a_D = p_1 d_1 + p_2 d_2 + p_3 d_3, \tag{11}$$
$$b_D = m_{12} d_1 d_2 + m_{13} d_1 d_3 + m_{23} d_2 d_3, \tag{12}$$
$$c_D = q\, d_1 d_2 d_3. \tag{13}$$

With

$$x_i = \frac{p_i d_i}{a_D}, \qquad x \in \Delta_2^\circ,$$

define

$$B_\beta(x) = \beta_{12} x_1 x_2 + \beta_{13} x_1 x_3 + \beta_{23} x_2 x_3. \tag{14}$$

Then

$$\frac{b_D}{a_D^2} = B_\beta(x), \qquad \frac{c_D}{a_D^3} = \kappa x_1 x_2 x_3. \tag{15}$$

4.1. The Fixed-Cubic Matignon Boundary

The fixed-polynomial component is a fractional Routh–Hurwitz/root-location problem; related coefficient-space criteria are established in the literature [8,9]. We state the exact form required for the diagonal-orbit elimination.

Lemma 1 (boundary value for a positive cubic)**.** *Let*

$$p(\lambda) = \lambda^3 + a\lambda^2 + b\lambda + c, \qquad a, b, c > 0,$$

and let

$$\frac{\pi}{3} < \theta < \frac{\pi}{2}.$$

Set

$$u = \cos\theta, \qquad K = 1 - 4u^2 > 0,$$

and define

$$r_\theta(a,b) = \frac{ua + \sqrt{u^2a^2 + Kb}}{K}, \qquad H_\theta(a,b) = r_\theta(a,b)^2\big(a + 2ur_\theta(a,b)\big). \tag{16}$$

Then

$$|\arg\lambda_j| > \theta \quad \forall j \quad \Longleftrightarrow \quad c < H_\theta(a,b). \tag{17}$$

Equality places one conjugate pair exactly on the rays $|\arg\lambda| = \theta$*. Moreover,*

$$H_\theta(sa, s^2b) = s^3H_\theta(a,b) \qquad (s > 0).$$

Proof. A polynomial with positive coefficients has no positive real zero, so loss of Matignon stability can occur only through a conjugate pair crossing a boundary ray. Put $\lambda = re^{i\theta}$, $r > 0$, in $p(\lambda) = 0$. Dividing the imaginary part by $r\sin\theta$ gives

$$r^2\frac{\sin 3\theta}{\sin\theta} + ar\frac{\sin 2\theta}{\sin\theta} + b = 0.$$

Using

$$\frac{\sin 3\theta}{\sin\theta} = 4u^2 - 1 = -K, \qquad \frac{\sin 2\theta}{\sin\theta} = 2u,$$

we obtain

$$Kr^2 - 2uar - b = 0,$$

whose unique positive solution is $r = r_\theta(a,b)$. At this solution the cubic has the explicit factorization

$$p(\lambda) = \big(\lambda^2 - 2ur\lambda + r^2\big)\big(\lambda + a + 2ur\big).$$

Indeed, $(1 - 4u^2)r^2 - 2uar - b = 0$ and $c = r^2(a + 2ur)$, so its roots are the boundary conjugate pair $re^{\pm i\theta}$ and the strictly negative third root $-a - 2ur < 0$. The real part then gives

$$c = r^2(a + 2ur) = H_\theta(a,b).$$

For $c > 0$ sufficiently small, the roots consist of a small negative real root and a perturbation of the roots of $\lambda^2 + a\lambda + b$, hence are inside the Matignon region. For $c \to \infty$, the roots approach the cube-root directions of $-c$, including arguments $\pm\pi/3$, which violate $\theta > \pi/3$. Since there is only one boundary value, Equation (17) follows. Homogeneity is immediate from Equation (16). □

For $2/3 < \alpha < 1$, put $\theta = \alpha\pi/2$ and define

$$h_\alpha(b) := H_\theta(1, b). \tag{18}$$

Equivalently,

$$r_\alpha(b) = \frac{u + \sqrt{u^2 + (1-4u^2)b}}{1-4u^2}, \qquad h_\alpha(b) = r_\alpha(b)^2\big(1 + 2u r_\alpha(b)\big), \tag{19}$$

where $u = \cos(\alpha\pi/2)$.

4.2. Exact Elimination of the Positive-Diagonal Orbit

Define

$$T_\alpha(\beta) = \min_{x \in \Delta_2^\circ} \frac{h_\alpha(B_\beta(x))}{x_1 x_2 x_3}. \tag{20}$$

For $2/3 < \alpha < 1$, writing $u = \cos(\alpha\pi/2) > 0$ and $K = 1 - 4u^2 > 0$ yields

$$h_\alpha(0) = \left(\frac{2u}{K}\right)^2 \left(1 + \frac{4u^2}{K}\right) > 0.$$

Since $B_\beta \geq 0$, the objective tends to $+\infty$ whenever $x \to \partial\Delta_2$, because its denominator $x_1 x_2 x_3 \to 0$. At the endpoint $\alpha = 1$, $h_1(b) = b$ and $h_1(0) = 0$, but instead

$$\frac{B_\beta(x)}{x_1 x_2 x_3} = \frac{\beta_{23}}{x_1} + \frac{\beta_{13}}{x_2} + \frac{\beta_{12}}{x_3} \longrightarrow +\infty \quad (x \to \partial\Delta_2),$$

for positive β. Thus the minimum is attained at an interior point for every $2/3 < \alpha \leq 1$.

Theorem 2 (exact three-dimensional Matignon D-stability)**.** *Let $-A$ be a strict P-matrix and let $2/3 < \alpha < 1$. Then*

$$\boxed{A \in \mathcal{F}_\alpha^{(3)} \iff \kappa < T_\alpha(\beta).} \tag{21}$$

Proof. For a positive diagonal D, let $s = a_D$ and let $x \in \Delta_2^\circ$ be its normalized orbit coordinate. Equations (15) give

$$a_D = s, \qquad b_D = s^2 B_\beta(x), \qquad c_D = s^3 \kappa x_1 x_2 x_3.$$

By Lemma 1 and homogeneity,

$$\operatorname{spec}(DA) \subset \Sigma_\alpha$$

is equivalent to

$$\kappa < \frac{h_\alpha(B_\beta(x))}{x_1 x_2 x_3}.$$

The normalization $D \mapsto x$ is onto the open simplex modulo common scale. Hence the inequality must hold for every $D \succ 0$ if and only if it holds for every $x \in \Delta_2^\circ$, which is exactly Equation (21). □

The theorem is an explicit low-dimensional solution of a generalized multiplicative D-stability problem. Abstract forbidden-boundary criteria for conic regions and complements are already known [6]; the step established in Theorem 2 is the complete elimination of the universal diagonal multiplier into the four invariants and the scalar threshold (20).

4.3. Interior and the Classical Cain Boundary

The strict-P hypothesis isolates the open full-dimensional principal-minor stratum to which Theorem 2 applies. The theorem does *not* classify matrices with one or more vanishing signed principal minors: on such boundary strata the normalization and threshold analysis may require separate limiting arguments. The following necessity result excludes these strata from the full-dimensional interior without asserting their complete classification.

Lemma 2 (principal-minor necessity). *If $A \in \mathcal{F}_\alpha^{(3)}$, then*

$$-a_{ii} \geq 0, \qquad \det A[\{i,j\}] \geq 0, \qquad -\det A > 0.$$

Consequently, every interior point of $\mathcal{F}_\alpha^{(3)}$ has $-A$ strict P.

Proof. If some $a_{ii} > 0$, let the other row scalings tend to zero; the limiting matrix has a positive real eigenvalue, and nearby positive diagonal scalings violate Matignon stability. If an order-two principal minor is negative, the same argument with the complementary row scaling tending to zero produces a 2×2 block with a positive real eigenvalue. Finally a real 3×3 Matignon-stable matrix has negative determinant, so $-\det A > 0$. Any zero order-one or order-two signed principal minor can be perturbed arbitrarily slightly to the forbidden sign, and therefore cannot occur at an interior point. □

Vanishing principal minors.

The excluded strata lie in the algebraic hypersurfaces $\{a_{ii} = 0\}$ and $\{\det A[I] = 0\}$; they are closed, have Lebesgue measure zero, and by Lemma 2 contain no interior point of $\mathcal{F}_\alpha^{(3)}$. The characterization of the full-dimensional interior is therefore unaffected. These strata are nevertheless not free of fractional-only matrices. Embedding the two-dimensional set (9),

$$A = \begin{pmatrix} 0 & b & 0 \\ c & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad bc < 0,$$

gives $\operatorname{spec}(DA) = \{\pm i\sqrt{-d_1 d_2 bc}, -d_3\}$ for every $D \succ 0$, so $A \in \mathcal{F}_\alpha^{(3)} \setminus \mathcal{D}_H^{(3)}$ for every $0 < \alpha < 1$, although $p_1 = p_2 = 0$. The four-invariant method does not extend directly to such points because its normalization degenerates: if $p_i = 0$, the simplex coordinate $x_i = p_i d_i / a_D$ vanishes identically and the invariants $\beta_I = m_I / \prod_{i \in I} p_i$ are undefined, while $m_I = 0$ removes the corresponding monomial from $c_{|I|}(D)$ in Equation (4). A classification of these lower-dimensional strata would require separate face-by-face limiting arguments and is not attempted here.

At the classical endpoint $\alpha = 1$,

$$h_1(b) = b,$$

so

$$T_1(\beta) = \min_{x \in \Delta_2^\circ} \left(\frac{\beta_{23}}{x_1} + \frac{\beta_{13}}{x_2} + \frac{\beta_{12}}{x_3} \right). \tag{22}$$

Cauchy–Schwarz gives the exact value

$$\boxed{T_1(\beta) = \left(\sqrt{\beta_{12}} + \sqrt{\beta_{13}} + \sqrt{\beta_{23}} \right)^2,} \tag{23}$$

which is Cain's strict-P real 3×3 D-stability threshold [1].

For $2/3 < \alpha < 1$, T_α lies strictly above this classical surface. Indeed, from

$$b = (1 - 4u^2)r^2 - 2ur$$

and Equation (19),

$$h_\alpha(b) - b = 4u^2r^2 + 2ur(r^2 + 1) > 0.$$

Therefore

$$\boxed{T_\alpha(\beta) > T_1(\beta) \qquad (2/3 < \alpha < 1).} \tag{24}$$

Corollary 1 (exact genuinely fractional band). *For* $2/3 < \alpha < 1$,

$$\operatorname{int} \mathcal{P}_\alpha^{(3)} = \{A : -A \text{ strict } P, \quad T_1(\beta) < \kappa < T_\alpha(\beta)\}. \tag{25}$$

For $0 < \alpha \leq 2/3$,

$$\operatorname{int} \mathcal{F}_\alpha^{(3)} = \{A : -A \text{ strict } P\}, \tag{26}$$

and

$$\operatorname{int} \mathcal{P}_\alpha^{(3)} = \{A : -A \text{ strict } P, \quad \kappa > T_1(\beta)\}. \tag{27}$$

Proof. For $2/3 < \alpha < 1$, Theorem 2 and Equation (23) give the two strict inequalities characterizing the full-dimensional interior. For $0 < \alpha \leq 2/3$, Kellogg's eigenvalue wedge theorem for P-matrices [7] implies, for every $D \succ 0$ and $\mu \in \operatorname{spec}(-DA)$, that

$$|\arg \mu| < \frac{2\pi}{3}, \qquad |\arg \lambda| = \pi - |\arg \mu| > \frac{\pi}{3} \geq \frac{\alpha\pi}{2}, \quad \lambda = -\mu.$$

The strict wedge inequality covers the endpoint $\alpha = 2/3$. Lemma 2 gives necessity for interior points, while Cain's threshold distinguishes the classical D-stable matrices from their complement. □

Remark 1 (fractional-only set versus its interior). *On the strict-*$P(-A)$ *stratum, for* $2/3 < \alpha < 1$, *the complete fractional-only* set *satisfies*

$$A \in \mathcal{P}_\alpha^{(3)} \quad \Longleftrightarrow \quad T_1(\beta) \leq \kappa < T_\alpha(\beta).$$

At $\kappa = T_1(\beta)$, *some positive diagonal scaling has a purely imaginary conjugate pair. Hence the orbit is not Hurwitz D-stable, although it remains strictly Matignon D-stable for every* $\alpha < 1$. *By contrast, the full-dimensional* interior *is characterized by* $T_1(\beta) < \kappa < T_\alpha(\beta)$, *as stated in Corollary 1.*

Figure 2 shows the exact threshold geometry on symmetric and asymmetric slices. The blue Matignon boundary remains strictly above the orange Cain boundary, and the region between them is precisely the all-positive-diagonal fractional-only band.

4.4. Minimum Dimension for a Robust Fractional-Only Class

The two-dimensional obstruction and the three-dimensional band give the dimension result.

Theorem 3 (minimum robust dimension). *For every* $0 < \alpha < 1$,

$$\boxed{\min\left\{n : \operatorname{int}\left(\mathcal{F}_\alpha^{(n)} \setminus \mathcal{D}_H^{(n)}\right) \neq \varnothing\right\} = 3.} \tag{28}$$

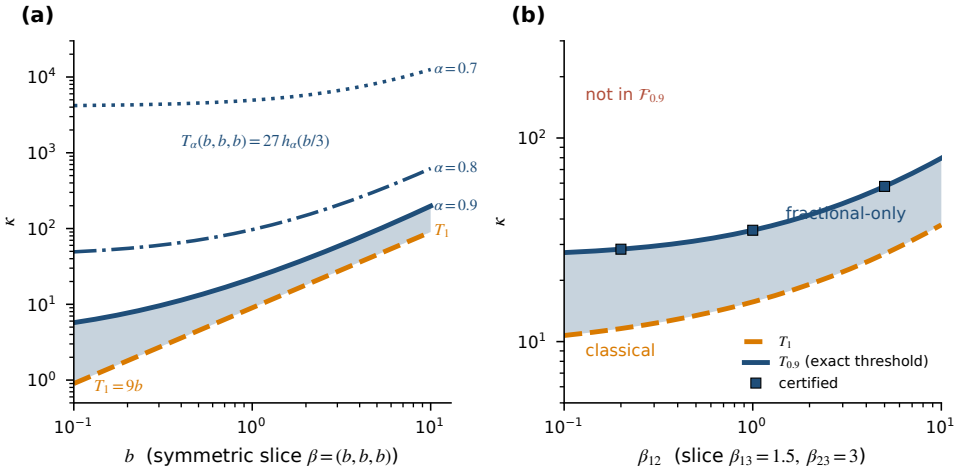


Figure 2. Exact positive-diagonal threshold geometry for real three-by-three matrices with strict-$P(-A)$ principal minors. Orange: classical Cain boundary $\kappa = T_1(\beta)$. Blue: fractional Matignon boundary $\kappa = T_\alpha(\beta)$ for the indicated order. The open interval $T_1 < \kappa < T_\alpha$ is fractionally stable for every positive diagonal multiplier but not classically Hurwitz D-stable. The numerical markers corroborate exact formulas and are not substitutes for the all-diagonal proof.

Proof. Dimension one is trivial and Theorem 1 gives empty full-dimensional interior in dimension two. For $2/3 < \alpha < 1$, Equation (24) makes the interval

$$T_1(\beta) < \kappa < T_\alpha(\beta)$$

nonempty for every positive β; any strict interior point is stable under small full-matrix perturbations. For $0 < \alpha \le 2/3$, choose any strict-$P(-A)$ matrix violating Cain's inequality $\kappa < T_1(\beta)$. Corollary 1 then places it in $\mathcal{P}_\alpha^{(3)}$, and strictness again gives a full-dimensional neighborhood. □

5. Geometry and Order Dependence of the Exact Threshold

The exact criterion in Theorem 2 reduces the complete positive-diagonal orbit to the scalar threshold $T_\alpha(\beta)$. We next show that this variational quantity has substantially more structure than a generic two-variable minimum.

5.1. Strict Convexity and a Unique Optimizer

Fix $2/3 < \alpha < 1$. Let

$$u = \cos\Big(\frac{\alpha\pi}{2}\Big), \qquad K_\alpha = 1 - 4u^2,$$

and recall from Equation (19) that $h_\alpha(b) = r^2(1 + 2ur)$, where r is the positive solution of

$$b = K_\alpha r^2 - 2ur.$$

Introduce the elasticity

$$E_\alpha(b) := \frac{b\,h'_\alpha(b)}{h_\alpha(b)}. \tag{29}$$

Lemma 3 (elasticity bounds)**.** *For every $b > 0$,*

$$0 < E_\alpha(b) < \frac{3}{2}, \qquad \frac{dE_\alpha}{db} > 0. \tag{30}$$

Proof. Using r as parameter,

$$\frac{db}{dr} = 2(K_\alpha r - u), \qquad \frac{dh_\alpha}{dr} = 2r(1+3ur),$$

and therefore

$$E_\alpha = \frac{(K_\alpha r - 2u)(1+3ur)}{(K_\alpha r - u)(1+2ur)}. \tag{31}$$

On $r > 2u/K_\alpha$, both factors

$$\frac{K_\alpha r - 2u}{K_\alpha r - u}, \qquad \frac{1+3ur}{1+2ur}$$

increase strictly with r. Their product increases from 0 to $3/2$. Since $b(r)$ is strictly increasing, Equation (30) follows. □

Parameterize the open simplex by logit variables

$$x_1 = \frac{e^{y_1}}{e^{y_1}+e^{y_2}+1}, \quad x_2 = \frac{e^{y_2}}{e^{y_1}+e^{y_2}+1}, \quad x_3 = \frac{1}{e^{y_1}+e^{y_2}+1}. \tag{32}$$

Set

$$L(y) = \log(e^{y_1}+e^{y_2}+1),$$

$$Q_\beta(y) = \log\big(\beta_{12}e^{y_1+y_2} + \beta_{13}e^{y_1} + \beta_{23}e^{y_2}\big),$$

and $\phi_\alpha(s) = \log h_\alpha(e^s)$. Then

$$G_{\alpha,\beta}(y) := \log\left[\frac{h_\alpha(B_\beta(x(y)))}{x_1x_2x_3}\right] = \phi_\alpha(Q_\beta - 2L) + 3L - y_1 - y_2. \tag{33}$$

Theorem 4 (strict threshold geometry). *For every $\beta \in \mathbb{R}^3_{>0}$ and $2/3 < \alpha < 1$, $G_{\alpha,\beta}$ is globally strictly convex on $\mathbb{R}^2$. Hence:*

1. *$T_\alpha(\beta)$ has a unique minimizer $x^*(\alpha,\beta) \in \Delta_2^\circ$;*
2. *the minimizer is nondegenerate;*
3. *x^* and T_α depend smoothly on (α,β).*

Proof. Put $s = Q_\beta - 2L$. Since

$$\phi'_\alpha(\log b) = E_\alpha(b), \qquad \phi''_\alpha(\log b) = b\,E'_\alpha(b),$$

Lemma 3 gives

$$\phi'_\alpha > 0, \qquad \phi''_\alpha > 0, \qquad 3 - 2\phi'_\alpha > 0.$$

Differentiating Equation (33),

$$\nabla^2 G_{\alpha,\beta} = \phi''_\alpha\,\nabla s \nabla s^\top + \phi'_\alpha \nabla^2 Q_\beta + (3 - 2\phi'_\alpha)\nabla^2 L. \tag{34}$$

Both Q_β and L are log-sum-exp functions of affine forms, so $\nabla^2 Q_\beta \succeq 0$; moreover $\nabla^2 L \succ 0$ because all three softmax weights are positive. Every coefficient in Equation (34) is nonnegative and the coefficient of $\nabla^2 L$ is strictly positive. Thus $\nabla^2 G_{\alpha,\beta} \succ 0$ everywhere.

As $\|y\| \to \infty$, $x(y)$ approaches the simplex boundary and $x_1x_2x_3 \to 0$, while $h_\alpha(B_\beta(x))$ remains positive. Hence $G_{\alpha,\beta}$ is coercive. A unique minimizer exists; positive definiteness gives nondegeneracy, and the implicit-function theorem gives smooth dependence. □

Two exact reductions follow immediately. If $\beta_{12} = \beta_{13} = \beta_{23} = b$, uniqueness and permutation symmetry imply $x^* = (1/3, 1/3, 1/3)$, so

$$T_\alpha(b, b, b) = 27 h_\alpha(b/3). \tag{35}$$

For $b = 1$, this becomes the cyclic-network boundary

$$T_\alpha(1, 1, 1) = 1 + R_3(\alpha)^3, \qquad R_3(\alpha) = \frac{\sin(\alpha\pi/2)}{\sin(\alpha\pi/2 - \pi/3)}, \tag{36}$$

recovering the structured slice studied by Siami [10]. If two β's are equal, the corresponding two simplex coordinates of x^* are equal, reducing the minimization exactly to one dimension.

5.2. Dependence on the Fractional Order

Theorem 5 (monotone widening and the classical limit)**.** *For fixed $\beta > 0$, $T_\alpha(\beta)$ is strictly decreasing in α on $2/3 < \alpha \le 1$. Let*

$$S = \sqrt{\beta_{12}} + \sqrt{\beta_{13}} + \sqrt{\beta_{23}}, \qquad G = \sqrt{\beta_{12}\beta_{13}\beta_{23}}.$$

As $\alpha \to 1^-$,

$$T_\alpha(\beta) = T_1(\beta) + C(\beta)(1 - \alpha) + O((1 - \alpha)^2), \tag{37}$$

where

$$C(\beta) = \pi\left(\frac{S^{5/2}}{\sqrt{G}} + S^{3/2}\sqrt{G}\right) > 0. \tag{38}$$

Proof. For fixed positive a, b, decreasing the boundary angle $\theta = \alpha\pi/2$ strictly enlarges the Matignon region. The unique coefficient value $H_\theta(a, b)$ placing a conjugate pair on the boundary therefore increases strictly as θ decreases. Hence $h_{\alpha_1}(b) > h_{\alpha_2}(b)$ for $\alpha_1 < \alpha_2$. Let x^* be the attained minimizer of the α_1 objective (whose existence follows from the coercivity established below Equation (20)). Then strictness is preserved under minimization by the explicit comparison

$$T_{\alpha_1}(\beta) = \frac{h_{\alpha_1}(B_\beta(x^*))}{x_1^* x_2^* x_3^*} > \frac{h_{\alpha_2}(B_\beta(x^*))}{x_1^* x_2^* x_3^*} \ge T_{\alpha_2}(\beta).$$

Taking $\alpha_2 = 1$, where $h_1(b) = b$, proves $T_\alpha > T_1$ by the same argument.

For the rate, $u = \cos(\alpha\pi/2) = \frac{\pi}{2}(1 - \alpha) + O((1 - \alpha)^3)$. Implicit differentiation of

$$(1 - 4u^2)r^2 - 2ur - b = 0$$

at $u = 0$ gives $\partial r/\partial u = 1$. Consequently

$$h_\alpha(b) = b + \pi(1 - \alpha)\sqrt{b}(1 + b) + O((1 - \alpha)^2). \tag{39}$$

At $\alpha = 1$, the unique simplex minimizer is

$$x_1^* = \frac{\sqrt{\beta_{23}}}{S}, \quad x_2^* = \frac{\sqrt{\beta_{13}}}{S}, \quad x_3^* = \frac{\sqrt{\beta_{12}}}{S},$$

for which

$$x_1^* x_2^* x_3^* = \frac{G}{S^3}, \qquad B_\beta(x^*) = \frac{G}{S}.$$

The envelope theorem applied to the nondegenerate minimum and Equation (39) yields Equations (37)–(38). □

Theorem 5 quantifies the part of the order dependence that is needed below: the fractional-only band widens strictly as α decreases and closes linearly at the classical endpoint. Together with the exact symmetric reductions above, this supplies a smooth and uniquely optimized threshold geometry without adding further parameters to the exact threshold classification.

6. A Nonlinear Caputo Double-Allee Ecological Realization

We now ask whether the open matrix band in Equation (25) can be realized by a biologically standard nonlinear Kolmogorov system rather than by an abstract matrix construction. The answer depends on the interaction architecture. The first natural model fails for a structural reason; an intraguild-predation architecture removes precisely that obstruction.

6.1. Positive-Equilibrium Orbit Equivalence

Consider a commensurate Caputo Kolmogorov system

$$ {}^{C}D_t^{\alpha} x_i = x_i F_i(x), \qquad i = 1, \dots, n, \tag{40} $$

and let $x^* \gg 0$ be an equilibrium. Put $B = DF(x^*)$.

Lemma 4 (Kolmogorov orbit equivalence). *At a positive equilibrium,*

$$ J(x^*) = \operatorname{diag}(x^*) B. \tag{41} $$

Consequently,

$$ J(x^*) \in \mathcal{F}_{\alpha}^{(n)} \iff B \in \mathcal{F}_{\alpha}^{(n)}, \qquad J(x^*) \in \mathcal{D}_{H}^{(n)} \iff B \in \mathcal{D}_{H}^{(n)}. $$

Proof. For $G_i(x) = x_i F_i(x)$,

$$ \partial_j G_i = \delta_{ij} F_i + x_i \partial_j F_i. $$

At x^*, $F_i(x^*) = 0$, which gives Equation (41). Since $D \mapsto D \operatorname{diag}(x^*)$ is a bijection of the positive diagonal cone onto itself, the two positive-diagonal spectral orbits coincide. □

This factorization is standard in Kolmogorov-system stability theory and is used here only as a bridge [15,16].

6.2. A Structural No-Go for Two Competing Consumers

Begin with a double-Allee basal prey consumed by two competitors:

$$ \begin{aligned} {}^{C}D_t^{\alpha} x &= x[g_{\mathrm{DA}}(x) - q_1 y - q_2 z], \\ {}^{C}D_t^{\alpha} y &= y[e_1 q_1 x - \mu_1 - c_{11} y - c_{12} z], \\ {}^{C}D_t^{\alpha} z &= z[e_2 q_2 x - \mu_2 - c_{21} y - c_{22} z], \end{aligned} \tag{42} $$

with positive parameters. At a positive equilibrium let $s = -g'_{\mathrm{DA}}(X) > 0$. The reduced matrix is

$$ B_{\mathrm{comp}} = \begin{pmatrix} -s & -q_1 & -q_2 \\ e_1 q_1 & -c_{11} & -c_{12} \\ e_2 q_2 & -c_{21} & -c_{22} \end{pmatrix}. \tag{43} $$

On its strict-P stratum,

$$\beta_{12} = 1 + \frac{e_1 q_1^2}{s c_{11}}, \qquad \beta_{13} = 1 + \frac{e_2 q_2^2}{s c_{22}}, \qquad \beta_{23} = 1 - \frac{c_{12} c_{21}}{c_{11} c_{22}} > 0, \tag{44}$$

and the normalized directed three-cycle contribution is

$$L_3 = \frac{q_1 q_2 (c_{12} e_2 + c_{21} e_1)}{s c_{11} c_{22}} > 0. \tag{45}$$

The determinant identity

$$\kappa = \beta_{12} + \beta_{13} + \beta_{23} - 2 - L_3 \tag{46}$$

then yields the following obstruction.

Theorem 6 (competitive-architecture no-go). *Every strict-P point of the model in Equation (42) is classically Hurwitz D-stable. In particular, the architecture cannot realize an open set satisfying*

$$T_1(\beta) < \kappa < T_\alpha(\beta).$$

Proof. From Equations (45)–(46),

$$\kappa < \beta_{12} + \beta_{13} + \beta_{23} - 2 < \beta_{12} + \beta_{13} + \beta_{23}.$$

Because all $\beta_{ij} > 0$,

$$T_1(\beta) = \left(\sqrt{\beta_{12}} + \sqrt{\beta_{13}} + \sqrt{\beta_{23}}\right)^2 > \beta_{12} + \beta_{13} + \beta_{23}.$$

Thus $\kappa < T_1(\beta)$, which is exactly Cain's strict-P D-stability inequality. □

The failure is therefore architectural rather than parametric: adding more parameter search cannot reverse the sign in Equation (45).

6.3. Intraguild Predation and Exact Invariant Access

Intraguild predation/omnivory is a standard three-species motif [17–19]. We therefore replace direct competition between the consumers by:

$$\begin{aligned} {}^C D_t^\alpha x &= x[g_{\mathrm{DA}}(x) - q_1 y - q_2 z], \\ {}^C D_t^\alpha y &= y[e_1 q_1 x - \mu_1 - c_1 y - h z], \\ {}^C D_t^\alpha z &= z[e_2 q_2 x + e_3 h y - \mu_2 - c_2 z], \end{aligned} \tag{47}$$

where

$$g_{\mathrm{DA}}(x) = \frac{r}{x + a}\left(1 - \frac{x}{K}\right)(x - m), \qquad 0 < m < K, \tag{48}$$

This is the standard multiplicative double-Allee growth form used in published predator–prey models: the factor $(x - m)$ supplies the threshold mechanism while the density-dependent intrinsic-growth factor $x/(x + a)$ supplies the second Allee mechanism [20, 21,23]. All displayed parameters are positive. Here x is the basal prey/resource, y the intermediate consumer or intraguild prey, and z the omnivorous top predator.

At a positive equilibrium (X, Y, Z),

$$g_{\mathrm{DA}}(X) = q_1 Y + q_2 Z, \tag{49}$$

$$c_1 Y + hZ = e_1 q_1 X - \mu_1, \tag{50}$$

$$-e_3 hY + c_2 Z = e_2 q_2 X - \mu_2. \tag{51}$$

Put

$$\Delta = c_1 c_2 + e_3 h^2, \quad A_1 = e_1 q_1 X - \mu_1, \quad A_2 = e_2 q_2 X - \mu_2.$$

Then

$$Y = \frac{c_2 A_1 - h A_2}{\Delta}, \qquad Z = \frac{e_3 h A_1 + c_1 A_2}{\Delta}. \tag{52}$$

Moreover

$$q_1 Y + q_2 Z = \chi X - \nu, \tag{53}$$

where

$$\chi = \frac{e_1 c_2 q_1^2 + e_2 c_1 q_2^2 + h q_1 q_2 (e_1 e_3 - e_2)}{\Delta}, \tag{54}$$

and

$$\nu = \frac{\mu_1 (q_1 c_2 + q_2 e_3 h) + \mu_2 (q_2 c_1 - q_1 h)}{\Delta}.$$

Thus X satisfies the degree-at-most-two equation

$$(r + K\chi)X^2 - [r(K + m) - K\chi a + K\nu]X + rKm - K\nu a = 0. \tag{55}$$

In the constructive regime used below $e_1 e_3 \geq e_2$, so $\chi > 0$ and Equation (55) is a genuine quadratic.

At coexistence set

$$s = -g'_{\mathrm{DA}}(X).$$

The reduced per-capita matrix is

$$B = \begin{pmatrix} -s & -q_1 & -q_2 \\ e_1 q_1 & -c_1 & -h \\ e_2 q_2 & e_3 h & -c_2 \end{pmatrix}. \tag{56}$$

Its order-two principal minors are

$$m_{12} = s c_1 + e_1 q_1^2, \quad m_{13} = s c_2 + e_2 q_2^2, \quad m_{23} = c_1 c_2 + e_3 h^2,$$

and a direct determinant calculation gives

$$-\det B = \Delta(s + \chi). \tag{57}$$

Hence the exact strict-P condition for this architecture is

$$s > 0, \qquad s + \chi > 0. \tag{58}$$

The four normalized invariants are

$$\beta_{12} = 1 + \frac{e_1 q_1^2}{s c_1}, \qquad \beta_{13} = 1 + \frac{e_2 q_2^2}{s c_2}, \quad \beta_{23} = 1 + \frac{e_3 h^2}{c_1 c_2}, \tag{59}$$

$$\kappa = \beta_{12} + \beta_{13} + \beta_{23} - 2 + \frac{h q_1 q_2 (e_1 e_3 - e_2)}{s c_1 c_2}. \tag{60}$$

Equivalently,

$$L_3 = \frac{hq_1q_2(e_2 - e_1e_3)}{sc_1c_2}. \tag{61}$$

Thus $e_1e_3 > e_2$ makes $L_3 < 0$, the sign mechanism missing from the competitive architecture.

Figure 3 displays the two motifs and the corresponding directed-cycle sign. The diagram is a theorem schematic; no numerical data enter the no-go or realization argument.

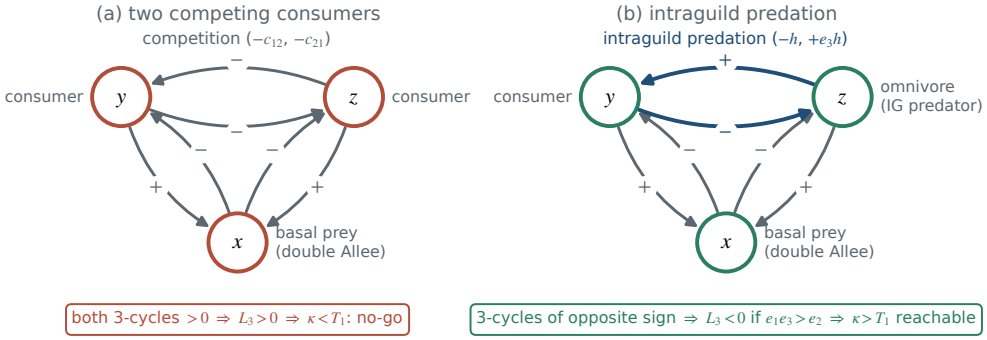


Figure 3. Two signed-interaction architectures for a double-Allee basal resource and two consumers. **(a)** Direct competition makes the signed three-cycle contribution $L_3 > 0$, forcing the normalized determinant below the classical Cain threshold $\kappa < T_1(\beta)$ for every strict-P coexistence equilibrium. **(b)** Intraguild predation reverses one oriented cycle and admits $L_3 < 0$ when $e_1e_3 > e_2$; the fractional-only interval $T_1 < \kappa < T_\alpha$ is then accessible. These graphs summarize an exact sign argument, not sampled ecological dynamics.

Theorem 7 (four-invariant IGP realization)**.** *Fix*

$$\beta_{12}, \beta_{13}, \beta_{23} > 1$$

and

$$\kappa > \beta_{12} + \beta_{13} + \beta_{23} - 2. \tag{62}$$

Then positive IGP parameters can be chosen so that the matrix in Equation (56) has exactly these four invariants. The construction may be made with

$$0 < e_1, e_2, e_3 < 1, \qquad e_1e_3 > e_2.$$

Proof. Let

$$A = \beta_{12} - 1, \quad B_0 = \beta_{13} - 1, \quad C = \beta_{23} - 1,$$

and

$$R = \kappa - (\beta_{12} + \beta_{13} + \beta_{23} - 2) > 0.$$

Put

$$\rho = \frac{R}{\sqrt{AB_0C}}, \qquad \tau = \frac{\rho + \sqrt{\rho^2 + 4}}{2} > 1,$$

so $\tau - \tau^{-1} = \rho$. Choose any $\eta \in (0, 1)$ and set

$$e_1 = e_3 = \eta, \qquad e_2 = \frac{\eta^2}{\tau^2}.$$

For arbitrary $s, c_1, c_2 > 0$, define

$$q_1 = \sqrt{\frac{Asc_1}{e_1}}, \qquad q_2 = \sqrt{\frac{B_0 sc_2}{e_2}}, \qquad h = \sqrt{\frac{Cc_1c_2}{e_3}}. \tag{63}$$

Equations (59) give the prescribed β's, while

$$\frac{hq_1q_2(e_1e_3 - e_2)}{sc_1c_2} = \sqrt{AB_0C}\left(\tau - \frac{1}{\tau}\right) = R.$$

Equation (60) then gives the prescribed κ. Since $\tau > 1$, $e_1e_3 > e_2$. □

6.4. Embedding the Invariants in the Double-Allee Model

Theorem 7 fixes a desired prey self-slope $s > 0$. Choose $a > 0$ and $0 < m < X < K$, and define

$$H_A = \frac{1}{K - X} - \frac{1}{X - m} + \frac{1}{X + a}. \tag{64}$$

With $C_A = (m + a)/[(X - m)(X + a)]$, one has $H_A = (K - X)^{-1} - C_A$; hence $H_A > 0$ precisely for

$$X < K < X + \frac{(X - m)(X + a)}{m + a}.$$

Set $Q = s/H_A$. For any $\xi \in (0, 1)$, prescribe

$$Y = \frac{\xi Q}{q_1}, \qquad Z = \frac{(1 - \xi)Q}{q_2},$$

so $q_1Y + q_2Z = Q$, and define

$$\mu_1 = e_1q_1X - Q\left(\frac{\xi c_1}{q_1} + \frac{(1 - \xi)h}{q_2}\right), \tag{65}$$

$$\mu_2 = e_2q_2X + Q\left(\frac{\xi e_3 h}{q_1} - \frac{(1 - \xi)c_2}{q_2}\right). \tag{66}$$

The first mortality is positive when

$$Q < Q_1 := \frac{e_1q_1X}{\xi c_1/q_1 + (1 - \xi)h/q_2}.$$

If $(1 - \xi)c_2/q_2 - \xi e_3 h/q_1 > 0$, require additionally

$$Q < Q_2 := \frac{e_2q_2X}{(1 - \xi)c_2/q_2 - \xi e_3 h/q_1};$$

otherwise $\mu_2 > 0$ automatically. Let $\bar{Q} = Q_1$ in the latter case and $\bar{Q} = \min(Q_1, Q_2)$ in the former. It is enough to choose K and r by

$$0 < K - X < \frac{1}{C_A + s/\bar{Q}}, \tag{67}$$

$$r = \frac{Q(X + a)}{(1 - X/K)(X - m)}. \tag{68}$$

Then $g_{\mathrm{DA}}(X) = Q = q_1Y + q_2Z$, while Equation (64) gives

$$g'_{\mathrm{DA}}(X) = Q\left[-\frac{1}{K - X} + \frac{1}{X - m} - \frac{1}{X + a}\right] = -QH_A = -s.$$

Thus the constructed positive equilibrium has exactly the reduced matrix of Theorem 7.

Theorem 8 (open genuinely fractional biological region). *For every fixed $0 < \alpha < 1$, there exists a nonempty open set in the full 14-dimensional positive biological parameter space of Equation (47) for which a positive coexistence equilibrium on the constructed smooth branch satisfies*

$$J \in \mathcal{F}_{\alpha}^{(3)} \setminus \mathcal{D}_{H}^{(3)}. \tag{69}$$

Proof. If $2/3 < \alpha < 1$, choose $\beta \in (1, \infty)^3$ and

$$T_1(\beta) < \kappa < T_\alpha(\beta).$$

Theorem 7 gives an IGP reduced matrix with these invariants, and Equations (64)–(68) embed it at a positive double-Allee coexistence equilibrium. Theorem 2 gives $B \in \mathcal{F}_{\alpha}^{(3)} \setminus \mathcal{D}_{H}^{(3)}$, and Lemma 4 transfers the statement to the actual Jacobian J.

If $0 < \alpha \leq 2/3$, choose $\beta > 1$ and any $\kappa > T_1(\beta)$ satisfying Equation (62). The construction is strict-P, so Kellogg's wedge implies $B \in \mathcal{F}_{\alpha}^{(3)}$, while Cain's inequality fails.

All feasibility and stability inequalities are strict. Moreover

$$-\det B = \kappa s c_1 c_2 > 0,$$

so the state Jacobian of the coexistence equations is nonsingular. The implicit-function theorem preserves the positive equilibrium on a smooth local branch under sufficiently small perturbations of all biological parameters, and continuity preserves the strict invariant inequalities. Hence, for fixed α, the admissible set is open in the full 14-dimensional biological parameter space. □

6.5. The Allee Threshold as a Monotone Invariant-Space Control

After eliminating Y and Z, the coexistence equation is

$$F(X, m) := g_{\mathrm{DA}}(X; m) - \chi X + \nu = 0. \tag{70}$$

Along a positive strict-P branch put

$$Q = g_{\mathrm{DA}}(X) = \chi X - \nu > 0, \qquad s = -g'_{\mathrm{DA}}(X) > 0.$$

By Equation (57),

$$s + \chi > 0.$$

Implicit differentiation gives the first exact monotonicity law:

$$\frac{dX}{dm} = -\frac{Q}{(X - m)(s + \chi)} < 0. \tag{71}$$

The slope s decreases as well. Set

$$d = X - m, \qquad \ell = K - X, \qquad w = X + a.$$

Then $H_A = \ell^{-1} - d^{-1} + w^{-1}$, and direct algebra gives

$$\partial_X H_A - H_A^2 = \frac{2(w - d)(\ell + w)}{d\ell w^2} > 0, \tag{72}$$

because $w - d = m + a > 0$. Differentiating $s = QH_A$ along Equation (70) yields

$$\frac{ds}{dm} = -\frac{Q}{d(s+\chi)}\left[Q\partial_X H_A + \frac{s}{d} + \chi\left(H_A + \frac{1}{d}\right)\right]. \tag{73}$$

Since strict-P gives $\chi > -s$, the bracket is strictly larger than

$$Q\partial_X H_A + \frac{s}{d} - s\left(H_A + \frac{1}{d}\right) = Q(\partial_X H_A - H_A^2) > 0.$$

Therefore

$$\boxed{\frac{dX}{dm} < 0, \qquad \frac{ds}{dm} < 0} \tag{74}$$

throughout the positive strict-P coexistence stratum. No sign assumption on χ is required.

Put $t = 1/s$ and define

$$A_0 = \frac{e_1 q_1^2}{c_1}, \quad B_0 = \frac{e_2 q_2^2}{c_2}, \quad C_0 = 1 + \frac{e_3 h^2}{c_1 c_2}, \quad E_0 = \frac{h q_1 q_2 (e_1 e_3 - e_2)}{c_1 c_2}.$$

Then

$$\beta_{12} = 1 + A_0 t, \quad \beta_{13} = 1 + B_0 t, \quad \beta_{23} = C_0, \quad \kappa = C_0 + (A_0 + B_0 + E_0)t, \tag{75}$$

and Equation (74) implies $dt/dm > 0$.

Along this path the classical gap is

$$\begin{aligned} G_1(t) &:= \kappa(t) - T_1(\beta(t)) \\ &= E_0 t - 2 - 2\sqrt{(1 + A_0 t)(1 + B_0 t)} \\ &\quad - 2\sqrt{C_0}\left[\sqrt{1 + A_0 t} + \sqrt{1 + B_0 t}\right]. \end{aligned} \tag{76}$$

Theorem 9 (unique transverse Cain crossing)**.** *If*

$$E_0 > 2\sqrt{A_0 B_0}, \tag{77}$$

then G_1 has a unique positive zero t_H. The crossing is transverse and satisfies

$$G_1'(t_H) \geq \frac{4 + 4\sqrt{C_0}}{t_H} > 0. \tag{78}$$

Consequently, because t increases strictly with m, the coexistence branch crosses the classical D-stability boundary at most once and does so transversally.

Proof. We have

$$G_1(0) = -4 - 4\sqrt{C_0} < 0,$$

whereas Equation (77) makes the asymptotic linear coefficient positive, so $G_1(t) \to +\infty$. Direct differentiation shows

$$\begin{aligned} G_1''(t) &= \frac{(A_0 - B_0)^2}{2[(1 + A_0 t)(1 + B_0 t)]^{3/2}} \\ &\quad + \sqrt{C_0}\left[\frac{A_0^2}{2(1 + A_0 t)^{3/2}} + \frac{B_0^2}{2(1 + B_0 t)^{3/2}}\right] > 0. \end{aligned} \tag{79}$$

Thus G_1 is strictly convex and has exactly one positive root. Convexity gives

$$G_1'(t_H) \geq \frac{G_1(t_H) - G_1(0)}{t_H},$$

which is Equation (78). □

The crossing can be prescribed. For any $t_0 > 0$, set

$$E_0 = \frac{2 + 2\sqrt{(1 + A_0 t_0)(1 + B_0 t_0)} + 2\sqrt{C_0}\left[\sqrt{1 + A_0 t_0} + \sqrt{1 + B_0 t_0}\right]}{t_0}. \tag{80}$$

Then $G_1(t_0) = 0$ and Equation (77) holds. Theorem 7 realizes this E_0 through a joint parameter choice, and the double-Allee embedding independently sets $s_0 = 1/t_0$ at a selected m_0. Hence, with every non-m model parameter fixed, there exists $\varepsilon > 0$ such that

$$m_0 - \varepsilon < m < m_0 \quad \Longrightarrow \quad J \in \mathcal{D}_H^{(3)},$$

whereas

$$m_0 < m < m_0 + \varepsilon \quad \Longrightarrow \quad J \in \mathcal{F}_\alpha^{(3)} \setminus \mathcal{D}_H^{(3)}. \tag{81}$$

For $\alpha > 2/3$, the right-hand implication follows from $T_\alpha > T_1$ and continuity; for $\alpha \leq 2/3$, strict-P itself gives fractional D-stability.

Figure 4 visualizes this mechanism for the certified realization used in Section 7. All non-m biological parameters are held fixed.

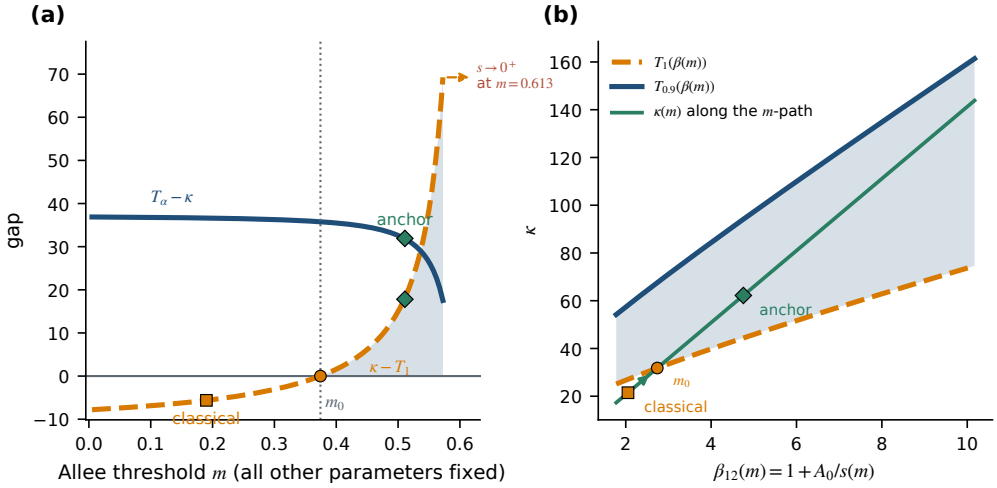


Figure 4. Effect of varying only the double-Allee threshold m on a fixed three-species intraguild-predation community at Caputo order $\alpha = 0.9$. **(a)** Invariant gaps versus m: Cain's classical Hurwitz D-stability gap $\kappa - T_1(\beta)$ crosses zero once near $m_0 = 0.374451$, while the fractional Matignon gap $T_{0.9}(\beta) - \kappa$ stays positive over the displayed interior interval. **(b)** The same parameterized path in an invariant-space projection: $\beta_{12}(m)$ and $\kappa(m)$ are compared with the exact Cain and fractional boundaries. Curves evaluate closed-form formulas; only the marked W2-A anchor and certified subinterval have rigorous interval semantics.

6.6. Exact Two- versus Three-Dimensional Ecological Contrast

For the corresponding two-species system

$${}^C D_t^\alpha x = x[g_{\mathrm{DA}}(x) - qy], \qquad {}^C D_t^\alpha y = y[px - d],$$

coexistence has $X = d/p$ and

$$B_2 = \begin{pmatrix} g'_{\mathrm{DA}}(X) & -q \\ p & 0 \end{pmatrix}.$$

Theorem 1 gives $B_2 \in \mathcal{F}_\alpha^{(2)}$ exactly when $g'_{\mathrm{DA}}(X) \le 0$, whereas classical Hurwitz D-stability requires the strict inequality. The genuinely fractional difference is therefore the codimension-one locus $g'_{\mathrm{DA}}(X) = 0$, equivalently

$$m_c = \frac{X^2 + 2aX - Ka}{K + a}. \tag{82}$$

Thus the two-species mechanism is exceptional, while Theorem 8 gives a nonempty open fractional-only region in three dimensions.

7. Certified Interior Realization

The preceding results are analytic and do not depend on numerical sampling. We nevertheless record one certified interior realization to show that the open ecological region can be reached with nondegenerate coexistence densities and comfortable separation from both the classical and fractional boundaries.

We use $\alpha = 0.9$ and the parameter set in Table 1. This realization was selected from a search over constructive embeddings only after the theorem package had been fixed; the search therefore chooses a readable interior witness rather than supplying evidence for the theorem.

Table 1. Canonical certified realization W2-A. All parameters are positive; the conversion efficiencies satisfy $0 < e_1, e_2, e_3 < 1$.

Parameter	Value	Parameter	Value	Parameter	Value
α	0.9	a	0.229	K	1.4913425743
r	7.8104480113	m	0.511	h	0.1404219270
q_1	0.5121684678	q_2	3.2338213736	c_1	0.065
c_2	0.0509	$e_1 = e_3$	0.773	e_2	0.01404592165
μ_1	0.3178535346	μ_2	0.1209173583		

At $m = 0.511$, interval evaluation gives a well-balanced positive coexistence point and a robust interior invariant margin:

$$(X, Y, Z) = (1,\ 0.7856328169\ldots,\ 0.1921819373\ldots), \tag{83}$$

$$s = 0.8231, \qquad \beta = (4.79,\ 4.506,\ 5.607), \qquad \kappa = 62.735, \tag{84}$$

$$T_1(\beta) = 44.6123995333883\ldots, \qquad T_{0.9}(\beta) = 94.5719074446105\ldots, \tag{85}$$

$$\kappa - T_1(\beta) \ge 18.1226004666, \qquad T_{0.9}(\beta) - \kappa \ge 31.8369074446. \tag{86}$$

The largest-to-smallest coexistence density ratio is approximately 5.20, and both certified margins place the point well inside the fractional-only band. The equilibrium is strictly biologically feasible ($X, Y, Z > 0$ and $m < X < K$); since the equilibrium equation has a simple root here, the implicit-function theorem preserves a nearby feasible equilibrium and the strict invariant margins under sufficiently small parameter perturbations. The size of this neighborhood is certified explicitly at the end of Section 7.1.

7.1. Interval Certification along the Allee Branch

The coexistence branch was enclosed with the outward-rounded `mpmath.iv` interval backend. Parameter-box and equilibrium evaluations used 50 decimal digits; the scalar equilibrium equation (70) was contracted by interval Newton evaluation, and the contracted state interval was propagated through Y, Z, s, β, κ and Cain's threshold. Crucially, uniqueness of the threshold minimizer alone would *not* validate a numerical lower bound. We therefore use 40-decimal-digit interval arithmetic to enclose a high-precision Newton candidate $\widehat{y}$ for the minimizer of the log-objective $G_{\alpha,\beta}$ in Equation (33). A subdivided box around $\widehat{y}$ is checked for a strictly positive interval lower bound $\underline{\lambda}$ on the Hessian's smallest eigenvalue (via Gershgorin or determinant-over-trace), and the gradient enclosure bounds the optimizer displacement by $\rho_y = \overline{\|\nabla G(\widehat{y})\|}/\underline{\lambda}$. The global strict-convexity theorem and the localization test place the true minimizer inside this box. An outward-rounded validated lower bound is then

$$\underline{T_\alpha} = \exp\Big(\underline{G_{\alpha,\beta}(\widehat{y})} - \overline{\|\nabla G_{\alpha,\beta}(\widehat{y})\|}\,\rho_y\Big) \le T_\alpha(\beta).$$

For interval parameter boxes, we take the componentwise lower corner $\underline{\beta}$, using monotonicity of T_α in each β_{ij}, and certify $\underline{T_\alpha}(\underline{\beta}) - \overline{\kappa} > 0$. As an independent check, simplex boundary-strip exclusion combined with interval branch-and-bound gives the same conclusion. The certificate thus requires both the analytic convexity result and explicit interval Hessian, gradient, and threshold enclosures; it is not inferred from an approximate optimizer.

A subdivision of the branch into 42 boxes was certified without failure on

$$0.394 \le m \le 0.578, \tag{87}$$

with the strict inequalities $\kappa - T_1 > 0$ and $T_{0.9} - \kappa > 0$ on every box. The smallest certified lower margins over the complete interval were 0.0332 and 1.342, respectively. The classical boundary was solved independently at

$$m_0 \approx 0.374451281242515, \tag{88}$$

The unrounded high-precision computation of the Cain-gap equation used 60 decimal digits and returned a residual below 10^{-59}; this residual pertains to the full internal approximation, not to the shortened printed number. This agrees with Theorem 9.

Certified 14-parameter neighborhood.

The same outward-rounded interval arithmetic certifies an explicit neighborhood of W2-A in the full parameter space. Let each of the 14 biological parameters vary independently in $[p_i(1-\delta),\ p_i(1+\delta)]$ about its value in Table 1, with e_1 and e_3 varied separately and $\alpha = 0.9$ fixed. For $\delta = 0.005$, an adaptive subdivision into 199 boxes certifies on every box, using first-order mean-value enclosures and a parametric interval Newton step for the prey equation, a unique enclosed root with $X, Y, Z > 0$, $m < X < K$ and $s > 0$ (hence $-B$ strict P), together with $\kappa - T_1(\beta) > 0$ and $T_{0.9}(\beta) - \kappa > 0$; the resulting enclosures are $X \in [0.981, 1.021]$, $Y \in [0.756, 0.813]$ and $Z \in [0.170, 0.212]$. The same certificate holds for $\delta = 0.0075$ with 28 802 boxes. Hence every parameter vector within 0.75% relative distance of W2-A in each coordinate has a feasible positive coexistence equilibrium in the open fractional-only band.

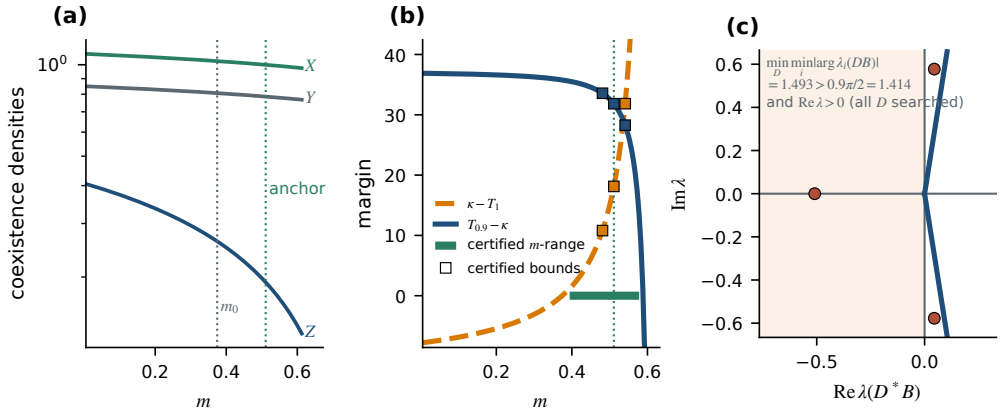


Figure 5. The three-species double-Allee intraguild-predation witness W2-A at order $\alpha = 0.9$ and anchor Allee parameter $m = 0.511$. **(a)** The positive coexistence densities X, Y, Z along the threshold branch (anchor approximately $(1, 0.786, 0.192)$). **(b)** Invariant gaps $\kappa - T_1(\beta)$ (non-Hurwitz D-stability) and $T_{0.9}(\beta) - \kappa$ (all-diagonal Matignon stability); only the marked lower bounds and green interval are interval-certified. **(c)** Eigenvalues at the positive diagonal ratio D_* selected by scale-invariant numerical angular search; the plotted spectrum is numerical corroboration, not the all-diagonal proof.

7.2. Scale-Invariant Spectral Corroboration

As an independent numerical check, we minimized the smallest eigenvalue angle over positive diagonal ratios after fixing the geometric mean of the diagonal entries to one. This normalization is essential: raw spectral abscissa is not scale invariant under common multiplication of D.

For the anchor, the most restrictive diagonal ratio found was approximately

$$D_* = \mathrm{diag}(0.1810,\ 2.0474,\ 2.6985), \qquad \mathrm{spec}(D_* B) = \{0.04494 \pm 0.57743i,\ -0.50930\},$$

and

$$\min_j |\arg \lambda_j(D_* B)| = 1.49312\ldots > 0.9\frac{\pi}{2},$$

while the conjugate pair has positive real part. Here $B = DF(u^*)$ is the reduced per-capita derivative and $J = \mathrm{diag}(u^*)B$ is the Jacobian of the *unscaled* nonlinear system. The search uses a selected member $D_* B$ of the positive-diagonal orbit, not necessarily J itself. Failure of Hurwitz D-stability means that $\exists D \succ 0$ with $\mathrm{spec}(DB) \not\subset \{\Re z < 0\}$, and does not assert that the unscaled J is unstable. The corresponding angular margin is about 0.0794 radians for $\alpha = 0.9$, whereas the classical $\alpha = 1$ margin is negative. This direct search corroborates both sides of the classification, but it is not used in the proof.

7.3. Full Nonlinear Time-Domain Case Study

We now integrate the complete nonlinear ecological system rather than its linearization. Let $u = (x, y, z)^\top$, let $u^* = (X, Y, Z)^\top$ be the anchor in Equation (83), and denote the unscaled nonlinear vector field in Equation (47) by $G(u)$. To realize exactly the orbit point above, set

$$R_* = D_* \,\mathrm{diag}(u^*)^{-1} = \mathrm{diag}(0.1810024,\ 2.6060229,\ 14.0412352) \tag{89}$$

and consider the positively row-scaled model

$${}^{C}D_t^{\alpha} u = R_* G(u). \tag{90}$$

The scaling changes only the three positive response rates, preserves the coexistence equilibrium and the interaction signs, and gives the exact Jacobian identity

$$R_* J = R_* \operatorname{diag}(u^*) B = D_* B.$$

The *unscaled* ecological model has Jacobian $J = \operatorname{diag}(u^*)B$; for the comparison we explicitly choose the positive response-rate configuration R_*, whose Jacobian is $R_* J = D_* B$. This selected row-scaled nonlinear model has a locally Matignon-stable equilibrium at $\alpha = 0.9$ and a classically unstable equilibrium at $\alpha = 1$. All-diagonal Matignon stability is a separate analytic statement.

Both integrations start from the identical relative perturbation

$$u(0) = u^* \odot \left(\mathbf{1} + 10^{-3}(1, -1, 1)^\top\right) = (1.001,\ 0.7848471840,\ 0.1923741192)^\top.$$

For $\alpha = 0.9$, we used the fractional Adams–Bashforth–Moulton PECE method [24–26] with $h = 0.01$ on $0 \le t \le 180$. Step halving gave maximum componentwise relative differences of 1.15×10^{-6} between $h = 0.04$ and 0.02, and 3.14×10^{-7} between $h = 0.02$ and 0.01, an observed refinement order of 1.875. The relative distance $\|u/u^* - \mathbf{1}\|_2$ decreased from 1.732×10^{-3} to 7.97×10^{-6}, and all three states remained positive.

Initial-condition perturbations.

Positivity near u^* holds for all times, not only locally in time. At every parameter vector of the certified box, $B \in \mathcal{F}_{0.9}^{(3)}$, so the Jacobian $RJ = R\operatorname{diag}(u^*)B$ is Matignon-stable for every positive diagonal response-rate matrix R, in particular for $R = I$ and $R = R_*$. The vector field is C^1 near u^*, so its remainder after linearization vanishes at u^* with local Lipschitz constant tending to zero; after a smooth modification away from a neighborhood of u^*, which does not affect solutions remaining in that neighborhood, the linearized asymptotic stability theorem for Caputo systems [27, Theorem 3.1] applies. Thus for every $\varepsilon > 0$ there is $\delta > 0$ such that $\|u(0) - u^*\| < \delta$ implies $\|u(t) - u^*\| \le \varepsilon$ for all $t \ge 0$ and $u(t) \to u^*$; choosing $\varepsilon < \min_i u_i^*$ keeps all three densities positive for all $t \ge 0$. The theorem does not quantify δ. As numerical evidence only, not a proof, we integrated 1024 random relative perturbations with $d(0) = \rho$ at each of 20 radii $0.01 \le \rho \le 0.9$ for the Table 1 parameters at $\alpha = 0.9$ (PECE $h = 0.01$ with refinement against $h = 0.02$, $0 \le t \le 600$). With R_*, every trajectory stayed positive and satisfied $d(600) \le \rho/10$ (in fact $d(600) \le 0.0027\rho$) for all $\rho \le 0.35$; at $\rho = 0.40$, 11 of 1024 trajectories did not return. For the unscaled model $R = I$ the corresponding radius was 0.15, with 87 failures at $\rho = 0.20$. Every non-returning trajectory approached the extinction state ($d(600) \approx \sqrt{3}$, all densities below 10^{-4} of their equilibrium values), as expected from the strong Allee threshold. These radii are finite-sample, finite-horizon estimates that depend on the response-rate configuration.

For $\alpha = 1$, an adaptive DOP853 integration with relative and absolute tolerances 10^{-11} and 10^{-13}, respectively, was stopped when the same relative distance first reached 0.2. This occurred at $t = 101.586$, consistent with the positive real part of the conjugate pair. Stopping at the declared radius keeps the comparison within a fixed local neighborhood; it is not a claim about the global fate of the classical trajectory.

Long-horizon nonlinear Caputo verification across the exact boundary order.

The four-invariant threshold defines the boundary order α_* by $T_{\alpha_*}(\beta) = \kappa$. The extremal positive diagonal obtained independently from the angular search satisfies, at the displayed precision,

$$\alpha_* = 0.9505501071\ldots = \frac{2}{\pi} \min_j |\arg \lambda_j(D_* B)|.$$

This is the critical order of the selected row-scaled Jacobian $R_*J = D_*B$. With the same W2-A model, positive response rates and initial perturbation, an additional sweep of the *full nonlinear* Caputo equations covered $0.7 \le \alpha \le 1$, including a dense grid with $\Delta\alpha = 0.001$ and additional points around α_*. The Adams–Bashforth–Moulton PECE runs used $h = 0.02$ and $h = 0.01$ for step refinement. The horizon was $t = 600$ away from the boundary and was extended to $t = 2000$ near it ($|\alpha - \alpha_*| < 0.02$). Integration was terminated when the relative distance $d(t) = \|u(t)/u^* - \mathbf{1}\|_2$ first reached the declared local-neighborhood radius 0.2; an independent DOP853 integration at $\alpha = 1$ reproduced the integer-order exit time.

Table 2 summarizes representative orders. For non-exiting trajectories, the late rate is the fitted finite-window slope of $\log d(t)$ over the last half of the integration, not an asymptotic exponential-decay exponent for fractional dynamics. Negative and positive fitted rates on either side of α_* corroborate the local spectral classification. The near-critical order $\alpha = 0.9506$ illustrates the danger of classifying solely by final versus initial distance: its late rate is positive, but $d(2000) = 1.322 \times 10^{-3} < d(0) = 1.732 \times 10^{-3}$. The unscaled configuration $R = I$ has a Hurwitz Jacobian and decayed at all tested orders.

Table 2. Long-horizon full nonlinear W2-A Caputo order sweep near the predicted order boundary $\alpha_* = 0.9505501071\ldots$, with fixed positive row scaling R_*. All runs share the same $10^{-3}(1, -1, 1)$ relative initial perturbation. The PECE method uses $h = 0.01$ with refinement against $h = 0.02$, terminating at radius $d = 0.2$. The late rate (units 10^{-3}) is a fitted finite-time slope of $\log d(t)$; a dash means that the trajectory exited and was stopped. Exit time is the first occurrence of $d = 0.2$. The numerical experiment corroborates but does not prove the analytic spectral boundary.

α	t_{end}	$d(t_{\text{end}})$	late rate (10^{-3})	exit time
0.9000	600	2.574×10^{-6}	-2.075	—
0.9450	2000	3.570×10^{-7}	-3.531	—
0.9500	2000	3.253×10^{-4}	-0.509	—
0.9506	2000	1.322×10^{-3}	$+0.046$	—
0.9510	2000	3.588×10^{-3}	$+0.417$	—
0.9520	2000	3.227×10^{-2}	$+1.347$	—
0.9530	2000	0.200	—	1955.0
0.9550	2000	0.200	—	1084.0
1.0000	600	0.200	—	101.6

Figure 7 displays trajectories and the late-rate sign change around the theoretically predicted order. In total, the two-stage sweep comprised 363 integrations. The end-versus-start distance diagnostic agreed with the linear spectral prediction in 339 of 340 dense runs; the sole near-critical exception has a positive late fitted rate but has not yet recovered its initial distance by $t = 2000$. This is finite-horizon numerical corroboration, not an independent proof of fractional asymptotic stability.

Finally, Figure 8 places the exact two-dimensional equality curve beside the three-dimensional open region. In panel (b), the region boundaries are traced by deterministic root finding applied to the exact formulas. The traced regions themselves are pointwise formula evaluations; only the green segment and marked anchor carry interval-certification semantics.

8. Discussion and Conclusions

The central distinction in this paper is between stability of one Jacobian and robustness of its complete positive-diagonal orbit. For a fixed commensurate Caputo linearization, the Matignon sector can admit a right-half-plane conjugate pair [4,5]. Here the stronger statement is that the *linearized Jacobian* remains Matignon-stable after every positive diag-

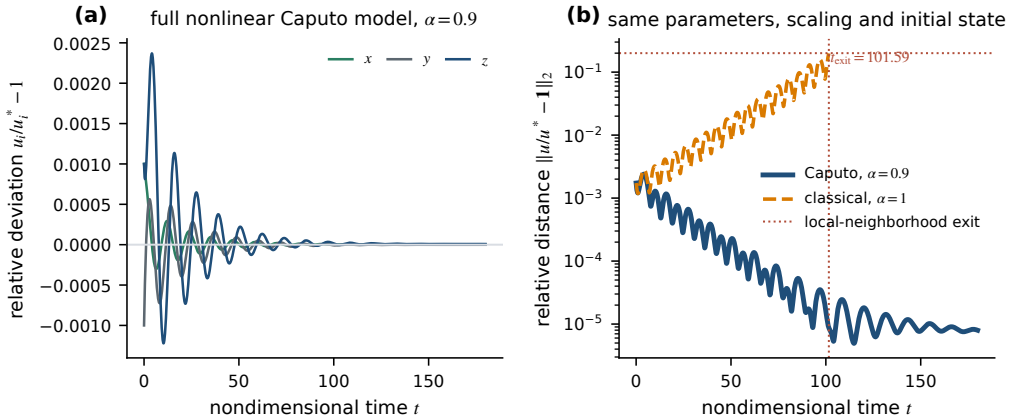


Figure 6. Full nonlinear time-domain comparison at the positive coexistence equilibrium of W2-A, with the selected positive response-rate scaling R_*. **(a)** Relative deviations $u_i(t)/u_i^* - 1$ for each species at Caputo order $\alpha = 0.9$, computed with PECE step $h = 0.01$, showing damped oscillations. **(b)** Relative Euclidean distance $\|u(t)/u^* - \mathbf{1}\|_2$ comparing $\alpha = 0.9$ against integer-order $\alpha = 1$, with identical parameters and initial state; the classical DOP853 trajectory ends upon reaching radius 0.2. The comparison concerns local dynamics, not global attraction.

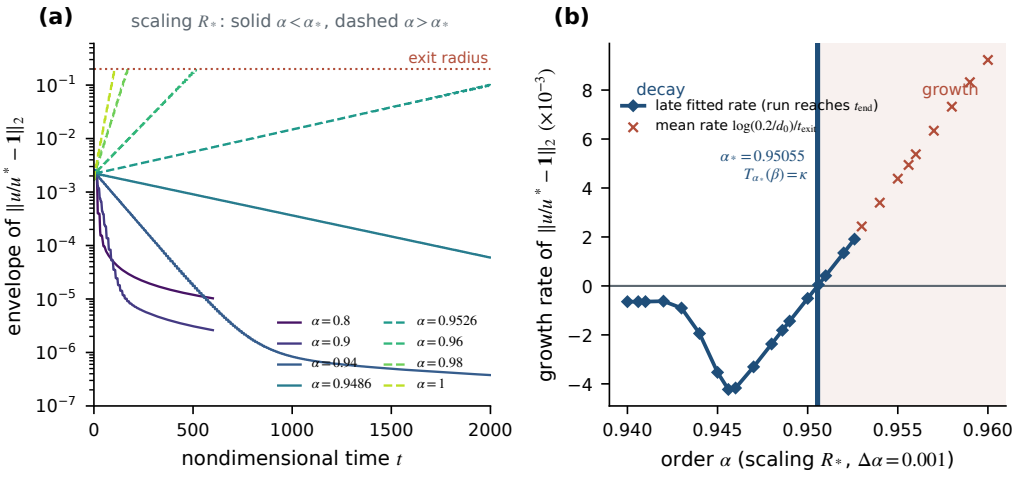


Figure 7. Full nonlinear three-species W2-A Caputo dynamics across the order boundary $\alpha_* = 0.9505501071\ldots$ for the fixed positive row-rate configuration R_*. **(a)** Running-window envelope of relative coexistence distance $d(t) = \|u(t)/u^* - \mathbf{1}\|_2$ versus nondimensional time for selected fractional orders on either side of the boundary; the dotted horizontal line marks the stopping radius $d = 0.2$. **(b)** Late fitted log-distance rate versus order α near the boundary. Diamonds are uninterrupted PECE integrations; crosses denote trajectories stopped at the exit radius and plot the mean rate $\log(0.2/d(0))/t_{\text{exit}}$ instead. The vertical line is the analytic threshold $T_{\alpha_*}(\beta) = \kappa$, not fitted to the trajectories. The simulations use PECE $h = 0.01$ with refinement at $h = 0.02$. The boundary follows from the theorem; the drawn trajectories and rates are finite-horizon numerical corroboration.

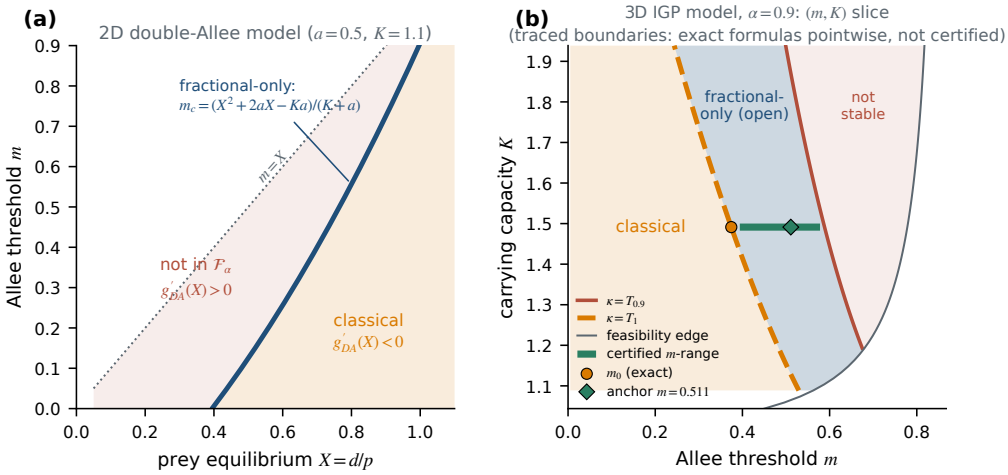


Figure 8. Geometric double-Allee contrast. **(a)** In the two-species model the fractional-only set in prey-equilibrium X versus Allee threshold m is the codimension-one curve $m = m_c(X)$. **(b)** For the three-species W2-A model at $\alpha = 0.9$, the (m, K) plane contains the classical Cain boundary $\kappa = T_1$, the fractional Matignon boundary $\kappa = T_{0.9}$, and the feasibility edge, enclosing an open fractional-only region. Curves and filled regions come from numerical evaluations of the exact formulas; only the indicated green interval is rigorously enclosed.

onal left scaling, even when the same orbit is not classically Hurwitz D-stable. Classical ecological D-stability is already a recognized robustness concept [16]; our contribution is the exact three-dimensional Matignon deformation of Cain's boundary and its nonlinear ecological realization.

The dimension statement must also be read precisely. Dimension three is not the first dimension in which fractional stabilization can occur. It is the minimum dimension in which

$$\mathcal{F}_\alpha^{(n)} \setminus \mathcal{D}_H^{(n)}$$

has nonempty full-dimensional interior. In two dimensions the difference class is confined to an equality locus; in three dimensions the exact interval $T_1(\beta) < \kappa < T_\alpha(\beta)$ is open. The strict-convexity result makes this boundary computationally and analytically well behaved: the threshold has a unique nondegenerate optimizer and varies smoothly with the invariant coordinates.

The ecological construction gives the invariant geometry a mechanistic interpretation without claiming novelty for IGP or Allee ecology. In the two-consumer architecture the directed three-cycle term has the sign that forces $\kappa < T_1$, so parameter tuning cannot reach the fractional-only band. In the IGP architecture one cycle changes sign and makes $L_3 < 0$ accessible. The resulting four-invariant right inverse, followed by the Double-Allee embedding, yields a nonempty open subset of the full 14-dimensional positive biological parameter space for each fixed α. Along that branch the Allee threshold has an exact matrix-geometric role: increasing m decreases $s = -g'_{\mathrm{DA}}(X)$, so $t = 1/s$ increases and can cross Cain's surface uniquely and transversally while all other biological parameters remain fixed.

The scope is local and spectral. Membership in $\mathcal{F}_\alpha$ concerns the linearized commensurate Caputo system; it does not prove global nonlinear stability, a Hopf bifurcation, or global uniqueness of positive coexistence. Biological feasibility is also a local property of the certified branch. At W2-A the coexistence equilibrium has $X, Y, Z > 0$ and $m < X < K$, the scalar root is simple, and both invariant inequalities have strict margins. The implicit-function theorem therefore preserves a nearby positive feasible equilibrium under sufficiently small perturbations of the 14 biological parameters; interval arithmetic

makes this explicit, certifying feasibility and the strict band for independent relative perturbations of up to 0.75% in all 14 parameters simultaneously, in addition to the segment $0.394 \leq m \leq 0.578$ at fixed remaining parameters (Section 7.1). For initial conditions, the linearized asymptotic stability theorem for Caputo systems [27] keeps every sufficiently small perturbation of u^* close to u^*, hence positive, for all $t \geq 0$, and gives convergence to u^*, for every positive response-rate configuration; the admissible radius is not quantified analytically. No global positivity or permanence conclusion follows from the local spectral certificate. The constructed equilibrium is a smooth local branch. The W2-A realization is a biologically feasible mathematical witness with certified margins, not an empirical calibration. The expanded nonlinear Caputo sweep consists of 363 integrations, including a dense order grid with step $\Delta\alpha = 0.001$, time-step refinement $h = 0.02$ to 0.01, and horizons up to $t = 2000$ near the exact threshold $\alpha_* = 0.9505501071\ldots$. Its late finite-window log-distance growth rate changes sign across the predicted boundary; the one apparent endpoint-distance disagreement immediately above the threshold is explained by long transients at finite observation times. For the unscaled, Hurwitz-stable ecological Jacobian, trajectories decay at all tested orders. Figure 7 illustrates these results; they corroborate the spectrum without constituting an independent proof of asymptotic stability. The direct diagonal search and full nonlinear time integration in Section 7 are reproducible corroboration rather than proof; the latter deliberately stops the unstable classical trajectory at a fixed local-neighborhood boundary.

In summary, the universal positive-diagonal quantifier for real 3×3 matrices on the strict-$P(-A)$ stratum reduces exactly to

$$\kappa < T_\alpha(\beta).$$

The threshold recovers Cain at $\alpha = 1$, lies strictly above it for $\alpha < 1$, and has a unique smooth optimizer. This yields the minimum full-dimensional robust dimension $n = 3$. A nonlinear Caputo Double-Allee IGP system then realizes an open subset of the resulting fractional-only class, while the competing-consumer architecture is structurally excluded and the Allee threshold supplies an exact crossing mechanism. For $n \geq 4$, the simplex reduction remains available but the number of nontrivial orbit invariants grows to $2^n - n - 1$; identifying higher-dimensional graph classes that retain comparable low-dimensional structure is the natural next problem.

Author Contributions: Conceptualization, I.A.; methodology, I.A.; formal analysis, I.A.; verification software, I.A.; writing—original draft, I.A.; writing—review and editing. I.A. has read and agreed to the submitted version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author

Acknowledgments: The author is thankful to the Deanship of Scientific Research at Islamic University of Madinah for supporting this work

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Cain, B.E. Real, 3×3, D-stable matrices. *Journal of Research of the National Bureau of Standards B* **1976**, *80B*, 75–77. https://doi.org/10.6028/jres.080B.013.
2. Hartfiel, D.J. Concerning the interior of the D-stable matrices. *Linear Algebra and its Applications* **1980**, *30*, 201–207. https://doi.org/10.1016/0024-3795(80)90195-0.
3. Kushel, O.Y. Unifying matrix stability concepts with a view to applications. *SIAM Review* **2019**, *61*, 643–729. https://doi.org/10.1137/18M119241X.

4. Matignon, D. Stability results for fractional differential equations with applications to control processing. In Proceedings of the Computational Engineering in Systems Applications: CESA'96 IMACS Multiconference, Symposium on Control, Optimization and Supervision, Lille, France, 1996; Vol. 2, pp. 963–968.
5. Brandibur, O.; Garrappa, R.; Kaslik, E. Stability of systems of fractional-order differential equations with Caputo derivatives. *Mathematics* **2021**, *9*, 914. https://doi.org/10.3390/math9080914.
6. Kushel, O.Y.; Pavani, R. The problem of generalized D-stability in unbounded LMI regions and its computational aspects. *Journal of Dynamics and Differential Equations* **2022**, *34*, 651–669. https://doi.org/10.1007/s10884-020-09891-y.
7. Kellogg, R.B. On complex eigenvalues of M and P matrices. *Numerische Mathematik* **1972**, *19*, 170–175. https://doi.org/10.1007/BF01402527.
8. Čermák, J.; Nechvátal, L. The Routh–Hurwitz conditions of fractional type in stability analysis of the Lorenz dynamical system. *Nonlinear Dynamics* **2017**, *87*, 939–954. https://doi.org/10.1007/s11071-016-3090-9.
9. Bourafa, S.; Abdelouahab, M.S.; Moussaoui, A. On some extended Routh–Hurwitz conditions for fractional-order autonomous systems of order $\alpha \in (0, 2)$. *Chaos, Solitons & Fractals* **2020**, *133*, 109623. https://doi.org/10.1016/j.chaos.2020.109623.
10. Siami, M. Stability and robustness analysis of commensurate fractional-order networks. *IEEE Transactions on Control of Network Systems* **2021**, *8*, 1261–1269. https://doi.org/10.1109/TCNS.2021.3061931.
11. Liu, J.; Li, R.; Huang, D. Stability, bifurcation and characteristics of chaos in a new commensurate and incommensurate fractional-order ecological system. *Mathematics and Computers in Simulation* **2025**, *236*, 248–269. https://doi.org/10.1016/j.matcom.2025.04.015.
12. Alomari, F.A.H.; Bahaa, G.M. Fractional-order analysis of a fear-induced ecoepidemiological predator–prey model with optimal control and bifurcation dynamics. *Scientific Reports* **2026**, *16*, 16130. https://doi.org/10.1038/s41598-026-52826-8.
13. Baghel, R.S. Hopf bifurcation and stability in a fractional-order age-structured predator-prey model with Allee effect and dual delays. *Chaos, Solitons & Fractals* **2026**, *205*, 117880. https://doi.org/10.1016/j.chaos.2026.117880.
14. Saha, A.; Pal, D.; Kesh, D.; Mukherjee, D. A fractional-order commensurate and incommensurate study on predator-prey system with Allee in predator and fear effect. *Fractional Calculus and Applied Analysis* **2026**, *29*, 1486–1515. https://doi.org/10.1007/s13540-026-00515-8.
15. Hou, Z.; Baigent, S. Global stability and repulsion in autonomous Kolmogorov systems. *Communications on Pure and Applied Analysis* **2015**, *14*, 1205–1238. https://doi.org/10.3934/cpaa.2015.14.1205.
16. Chen, C.; Wang, X.W.; Liu, Y.Y. Stability of ecological systems: A theoretical review. *Physics Reports* **2024**, *1088*, 1–41. https://doi.org/10.1016/j.physrep.2024.08.001.
17. Holt, R.D.; Polis, G.A. A theoretical framework for intraguild predation. *The American Naturalist* **1997**, *149*, 745–764. https://doi.org/10.1086/286018.
18. Bai, D.; Kang, Y.; Ruan, S.; Wang, L. Dynamics of an intraguild predation food web model with strong Allee effect in the basal prey. *Nonlinear Analysis: Real World Applications* **2021**, *58*, 103206. https://doi.org/10.1016/j.nonrwa.2020.103206.
19. Panja, P. Dynamics of a fractional order predator–prey model with intraguild predation. *International Journal of Modelling and Simulation* **2019**, *39*, 256–268. https://doi.org/10.1080/02286203.2019.1611311.
20. Alraddadi, I.; Ahmed, R.; Seol, Y. Bifurcation structure and chaos control in a discrete-time fractional predator–prey model with double Allee effect. *Fractal and Fractional* **2026**, *10*, 304. https://doi.org/10.3390/fractalfract10050304.
21. Mondal, C.; Mondal, R.; Kesh, D.; Mukherjee, D. Dynamics of predator-prey system with the consequences of double Allee effect in prey population. *Journal of Biological Physics* **2025**, *51*, 5. https://doi.org/10.1007/s10867-025-09670-0.
22. Kinkhabwala, A. Implications of network topology on stability. *PLOS ONE* **2015**, *10*, e0122150. https://doi.org/10.1371/journal.pone.0122150.
23. Pal, P.J.; Saha, T. Qualitative analysis of a predator–prey system with double Allee effect in prey. *Chaos, Solitons & Fractals* **2015**, *73*, 36–63. https://doi.org/10.1016/j.chaos.2014.12.007.
24. Garrappa, R. On linear stability of predictor–corrector algorithms for fractional differential equations. *International Journal of Computer Mathematics* **2010**, *87*, 2281–2290. https://doi.org/10.1080/00207160802624331.
25. Garrappa, R. Numerical solution of fractional differential equations: A survey and a software tutorial. *Mathematics* **2018**, *6*, 16. https://doi.org/10.3390/math6020016.
26. Diethelm, K. *The Analysis of Fractional Differential Equations*; Vol. 2004, *Lecture Notes in Mathematics*, Springer: Berlin/Heidelberg, Germany, 2010.
27. Cong, N.D.; Doan, T.S.; Siegmund, S.; Tuan, H.T. Linearized asymptotic stability for fractional differential equations. *Electronic Journal of Qualitative Theory of Differential Equations* **2016**, pp. 1–13. https://doi.org/10.14232/ejqtde.2016.1.39.